How Misleading Information Change Individuals' Perception: Fake News and Mixed Effects of Identities in Taiwan

1. Introduction

A growing body of literature reveals that fake news poses a serious threat to democracy. Fake news fosters epistemic cynicism, affective polarization, and pervasive inauthenticity, ultimately sowing chaos and obscuring the truth (McKay and Tenove 2021). Due to its impact, external actors such as China and Russia can exploit the vulnerabilities of democracy for the purpose of destabilizing targeted societies and manipulating their policy-making processes (Beauchamp-Mustafaga 2023; Bernal et al. 2020; Lin et al. 2022). Notably, social media, which serve as primary channels of communication, exponentially spread misinformation, thereby causing distorted voter perceptions and influencing electoral behaviors (Backes and Swab 2019; Howard et al. 2019).

However, how individual process misleading information or information and how it reconstructs their perceptions remain in puzzle. There are various mechanisms and evidence regarding the motivation for information processing. When people encounter new information, they interpret it based on their identities, and many studies are conducted with a focus on partisanship (Gaines et al. 2007). One of the effects of motivated reasoning is polarization which is driven by partisan-motivated reasoning. Although many studies provide evidence that confirmation bias and disconfirmation bias are main explanations of these effects (Charness and Dave 2017; Ho et al. 2017; Taber and Lodge 2006), the effects of polarization by motivated reasoning are a topic of ongoing research (Druckman and McGrath 2019; Jost et al. 2022). On the other hand, Bayesian Learning Model suggests that people pursue accuracy by correcting their belief while processing information (Guess and Coppock 2020; Hill 2017; Tappin,

Pennycook, and Rand 2020).

Since these two approaches provide different explanations on how individuals process new information, the results of process are different accordingly. Moreover, individuals possess overlapping identities that can interact or, in certain contexts, identity saliency can be changed (Unsworth and Fielding 2014). This suggests that identities other than partisanship may have a more prominent influence on motivated reasoning. Alternatively, identities such as gender and race could act as moderators for partisan motivated reasoning (Boyer et al., 2022). In situations of external intervention through disinformation or the presence of adversarial countries, fake news regarding specific topics allows individuals to speculate about the attribution of fake news. In such cases, instead of polarization, a "rally 'round the flag" effect can occur (Bauer and Wilson 2022).

What are the effects of mix-identity motivated reasoning? This study aims to evaluate the different effects of mix-identity motivated reasoning by fake news. The effects of mix-identity motivated reasoning can be profound. In the real world, where individuals possess multiple identities, the identities evoked by misinformation can also be diverse. This complexity can significantly influence how information is processed and accepted, potentially complicating efforts to correct or counteract false information. The saliency of identity influences the mechanisms through which people process information, and the effects of motivated reasoning vary accordingly. Thus, the question arises: When do dominant identities change to another, how do these identities interact with each other, and what are the effects of these interactions?

Taiwan is an attractive place to capture the effects of mix-identity motivated reasoning on misinformation because there exist two salient identities: partisanship and Taiwanese identity. Individuals in other regions can simultaneously hold nationalities and partisanship. However, in domestic political issues, partisanship tends to operate as the more salient identity compared

to nationality. In contrast, Taiwan has experienced conflicts between Mainlanders and indigenous ethnic groups since the establishment of the Kuomintang government in 1945, and after democratization, the basis of partisanship was ethnicity (Achen and Wang 2017). Furthermore, this ethnicity-based partisanship has diverged into Chinese identity and Taiwanese identity, and perspectives on cross-strait relations vary according to these identities. In situations where cross-strait relations are intensifying, nationality acts as the dominant identity in issues related to cross-strait relations, which are also prominent in Taiwan. Socioeconomic issues are another factor that divides partisanship in Taiwan. Achen and Wang (2017) identify issues such as reform versus stability, wealth distribution, and the trade-off between environmental protection and economic growth as the bases for issue-based partisanship.

The 2024 Taiwan presidential election presents an excellent case study for testing hypotheses related to mixed-identity motivated reasoning and the median voter's Bayesian updating. In the 2024 Taiwan presidential election, a third party emerged prominently, capturing a significant share of the votes. This indicates that a substantial number of voters withdrew their support from the established parties and supported the new party, thereby allowing us to assess the effects of the median voter's information processing. Furthermore, the presence of voters who continued to support their original parties even as the median voter shifted away suggests that these voters possess relatively strong partisanship, from which we can infer the effects of partisan motivated reasoning. Lastly, since cross-strait relations remained a salient issue amidst worsening international conditions in the 2024 Taiwan presidential election, the effects of nationality motivated reasoning can also be captured.

I designed my research in two parts. First, I collected data on fake news in Taiwan from the Taiwan FactCheck Center, covering the period between the presidential elections from 2020

to 2024. Based on this collected data, I conducted topic modeling to classify which identities each topic evokes among Taiwanese voters. When considering fake news, the degree to which people pay attention to the news varies depending on the issue's saliency. The less salient a topic is, the more likely people are to engage in Bayesian updating rather than motivated reasoning. To measure topic saliency, I calculate the Herfindahl-Hirschman Index (HHI) for the extracted topics. Secondly, using a fixed effects model, I analyze how voters with different partisanship and nationalities change their perceptions of the opponent party when exposed to fake news that evokes different identities. In this analysis, I also include perceptions of the Chinese Communist Party (CCP) to study the effects of China-related fake news.

My study offers significant insights into the impact of fake news on both political behavior and the cognitive warfare aspect of international relations. (Contribution: write after getting results)

To explain the effects of the fake news on people's perception, I explored potential dynamics through examining the effects and interactions of two salient identities. I find that (findings: write after getting results)

2. The Effect of Misleading Information

There are various mechanisms and evidence regarding the motivation for information processing. When people encounter new information, they interpret it based on their identities, such as partisanship (Gaines et al. 2007) or other social identities like gender and race (Boyer et al. 2022). One of the effects of motivated reasoning is polarization which is driven by partisan-motivated reasoning. Although many studies provide evidence that confirmation bias and disconfirmation bias are main explanations of these effects (Charness and Dave 2017; Ho et al. 2017; Taber and Lodge 2006), the effects of polarization by motivated reasoning are a topic of ongoing research (Druckman and McGrath 2019; Jost et al. 2022). On the other hand,

Bayesian Learning Model suggests that people pursue accuracy by correcting their belief while processing information (Guess and Coppock 2020; Hill 2017; Tappin, Pennycook, and Rand 2020). This mixed conclusion demonstrates the need for further research on the effects of information processing among individuals.

The study of misinformation is grounded on the achievements of cognitive psychology, extending across various domains, yet the puzzles identified in prior research remain unresolved. Numerous studies argue that the acceptance and processing of misinformation are influenced by motivated reasoning causing polarization. Prior exposure to fake news increases the perceived accuracy of fake news, inhibiting people's accuracy pursuing motive (Pennycook, Cannon, and Rand 2018), and repetition of such false information increases belief in the information, even when the misinformation contradicts prior knowledge (Pillai and Fazio 2021). Individuals are more inclined to accept misinformation aligning with their political beliefs and to dismiss information or misinformation that threatens their identity (Grady et al., 2021; Peterson & Iyengar, 2021; Vegetti & Mancosu, 2020).

However, individuals possess overlapping identities that can interact or, in certain contexts, identity saliency can be changed (Unsworth and Fielding 2014). This provides hypothesis that identity other than partisanship may exert a more salient influence on motivated reasoning. Consequently, the effects of misinformation can vary accordingly. In situations of external intervention through disinformation or the presence of adversarial countries, fake news regarding specific topics allows individuals to speculate about the attribution of fake news. In such cases, instead of polarization, a "rally 'round the flag" effect can occur (Bauer and Wilson 2022).

If the effects of motivated reasoning differ based on individuals' various identities, when does a specific identity become salient among overlapping identities? What are the interactive

effects among these identities? I argue that different topics of fake news will stimulate different identities. For example, if the fake news pertains to China-linked information, Taiwanese citizens might speculate that the source of the fake news is China evoking backlash effects (Bauer and Wilson 2022). In contrast, if the fake news is related to the food security system or the COVID-19 quarantine system, it might shift the salience of identity towards partisanship. Otherwise, people might seek accuracy of information. That said, it is necessary to clarify the topic of fake news before testing perceptional change of individuals.

3. Mixed Identity-Motivated Reasoning on the Fake News in Taiwan

People are not fully informed, and, in situations of incomplete information, they rely on prior beliefs to process contradictory information. In this process, individuals may seek accuracy, but influenced by their identity, they may also become attached to their beliefs through motivated reasoning. What determines when people engage in Bayesian updating versus when they resort to motivated reasoning? Furthermore, what is the effect of motivated reasoning in situations where two salient identities coexist?

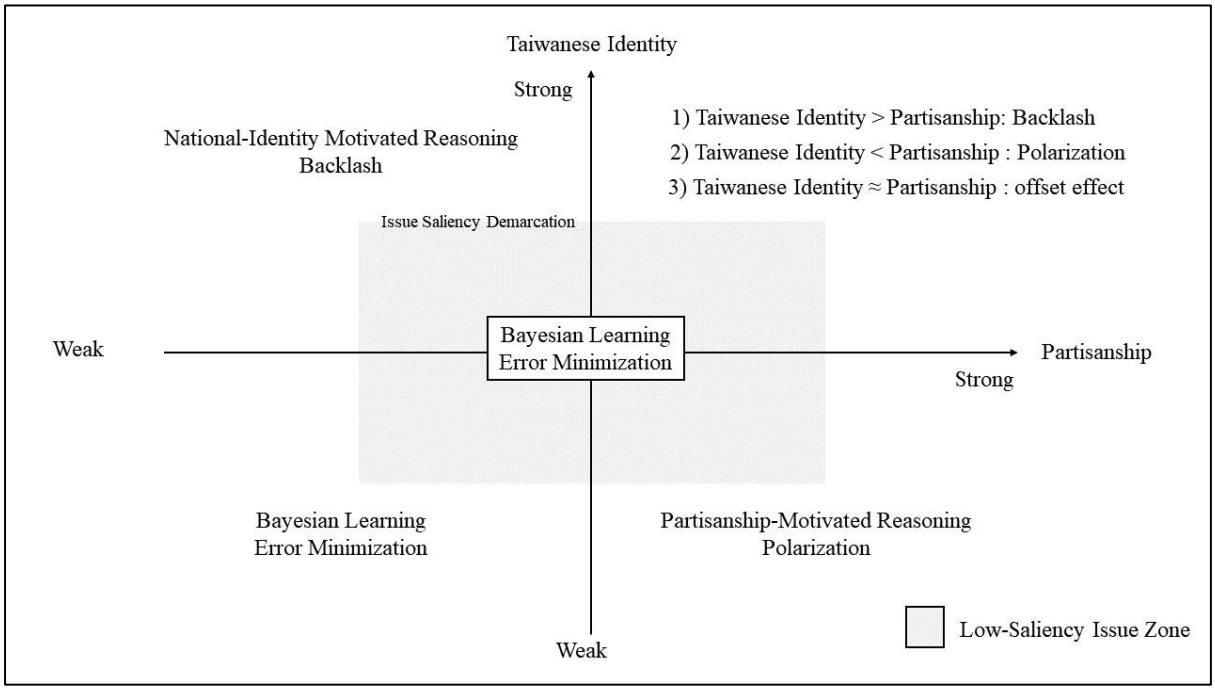
Taiwan provides a favorable context for understanding the interaction among identities and its influence on individuals' political perceptions. For Taiwanese people, not only partisanship but also nationality serve as lenses through which they view political issues. In this chapter, we will first present a theory on the mixed effects of multiple identities in information, followed by its application to Taiwanese society to formulate hypotheses.

1) Mixed Effects of multiple Identities in Information Processing

Figure 1 illustrates the mixed effects of two types of identity. Each quadrant is distinguished by the type of issue, with each topic triggering either a social identity or partisanship in individuals. Quadrants 2 and 4 represent cases where the topic of information

activates nationality and political identities, respectively. In these cases, individuals engage in motivated reasoning influenced by the activated identity. Conversely, Quadrant 3 corresponds to situations where the topic of information scarcely stimulates either identity, leading to Bayesian updating in people's information processing; as a result, people seek accuracy of the information rather than change their political perspectives. Lastly, Quadrant 1 represents cases where both identities are strongly triggered simultaneously. In this situation, the extent of the identities of those handling it becomes crucial. If nationality is a more dominant identity than partisanship, then people activate nationality-motivated reasoning in the information processing, and vice versa. Lastly, if the nationality and partisanship identities are similarly dominant, the effects of the two types of motivated reasoning will cancel each other out.

Figure 1. Interaction between Two Identities and the Effects of Motivated Reasoning



However, the type of topic alone is insufficient to explain the changing points in people's information processing. That is, for individuals to engage in motivated reasoning driven by a specific identity, the issue must be sufficiently salient to them. In Figure 1, I have denoted this as the issue saliency demarcation. The area marked in light gray represents the zone where

Bayesian updating occurs. In the case of Quadrant 3, where the issue type itself does not evoke specific identities, Bayesian updating takes place; however, the gray area spanning across the quadrants signifies areas where the issue does not strongly evoke an identity enough for motivated reasoning.

2) Two Salient Identities in Taiwan

Taiwan is broadly characterized by two main political parties, the Kuomintang (KMT) and the Democratic Progressive Party (DPP), each following a distinct approach toward relations with China. Broadly speaking, the Kuomintang (KMT) advocates for reunification with mainland China, while the Democratic Progressive Party (DPP) favors independence for Taiwan.

Following democratization, the KMT abandoned its Three Noes Policy,[1] which stipulated no negotiation or exchange with mainland China, and instead pursued cooperation and dialogue. Subsequently, Taiwan needed to redefine its relationship with mainland China for the sake of exchanges. This led to the establishment of the 1992 Consensus. The 1992 Consensus outlines cross-strait relations with the principle of "one China, different interpretations" as its core. In essence, while agreeing to aspire towards long-term reunification, each side interprets who holds the legitimate representation of "one China" differently. However, the PRC's stance asserts that the legitimate representative of one China, including Taiwan, is the mainland China.

The KMT's pro-China approach became a target of attack for the DPP. Founded in 1986, the DPP advocated for Taiwan independence, explicitly stating it in its party charter, thus supporting the idea of dual recognition. However, over time, in order to appeal to median voters who supported the status quo, the DPP softened its pro-independence stance by adopting a

[1] No contact, no negotiation, no compromise

more moderate form of separatism, namely the maintenance of the status quo. The DPP opposes accepting the 1992 Consensus, arguing that it does not constitute official government-level talks but rather mere verbal exchanges of opinions.

Backlash Hypothesis

The pro-China and pro-independence stances in Taiwan are closely intertwined with the Taiwanese's social identity, primarily demonstrated through partisan alignment rooted in national identity (Sota and Gethin 2021). Individuals who identify themselves as both Chinese and Taiwanese typically favor integration with mainland China and exhibit allegiance to the Kuomintang (KMT). Conversely, those embracing a Taiwanese identity advocate for disintegration from mainland China and tend to support the Democratic Progressive Party (DPP).

In previous presidential elections, partisan sorting has been driven by national identity, making cross-strait relations a salient issue in the elections. Of note is the continuous rise in the identification of oneself as Taiwanese, while the proportion of individuals identifying with a mixed identity as both Taiwanese and Chinese is gradually decreasing.[2] This trend signifies the salience of social identity as Taiwanese, driven by a sense of threat perception from mainland China and a desire to maintain political autonomy for Taiwan.

As Bauer and Wilson (2022) provides, in a situation where Taiwanese identity is increasing and the perception of threat from China is strengthening, we can formulate the following hypothesis regarding Taiwanese people's motivated reasoning toward fake news.

[2] Over time, the proportion of Taiwanese identifying as both Taiwanese and mainland Chinese has been decreasing, while the identity of distinguishing themselves as Taiwanese, separate from mainland Chinese, has steadily increased. As of 2023, 32% of people in Taiwan identify as both Taiwanese and Chinese, and 61.7% identify solely as Taiwanese, indicating a growing divide between these identities over time. (Election Study Center, National Chengchi University, 2024).

- H1: Exposure to fake news associated with China is likely to worsen Taiwanese citizens' perception of the CCP.
- H1a: The support for the KMT among Taiwanese citizens is expected to decline as they encounter fake news associated with China.
- H1b: The support for the DPP among Taiwanese citizens is likely to increase as they encounter fake news associated with China.

Polarization Hypothesis

How do people in Taiwan handle fake news on different topics? And how does the effect of such misguided information vary depending on issue saliency? The 2024 Taiwan presidential election showed a different trend compared to the elections in 2016 and 2020. First, there was the Kuomintang (KMT) strategy targeting Taiwanese identity. KMT candidate Hou Yu-ih, a former police officer with roots in Taiwan, had no connections with China and had never visited China. During the presidential debates, he made efforts to appeal to Taiwanese identity by using the Taiwanese indigenous language. This strategy was a response to the backlash against the KMT's pro-China stance, as seen in the 2020 presidential election with candidate Han Kuo-yu, who faced accusations of being pro-China due to his Hong Kong security law stance. The KMT aimed to dilute its pro-China image and emphasize Taiwanese representation.

Secondly, there was the Kuomintang's distancing from China. Candidate Hou presented a 3D strategy of deterrence, dialogue, and de-escalation during his visit to the United States in September 2023. He explicitly mentioned deterrence, signaling a willingness to cooperate with China while firmly addressing any threats. Additionally, in an official statement regarding the 1992 Consensus on July 3, 2023, he mentioned "accepting one, rejecting two."[3] Hou expressed

[3] Chen, Yi-yun, Zhou, Yu-xiang, & Wu, Hong-xun. 2023. "Hou Yu-ih： Embracing the 1992 Consensus in Accordance with

acceptance of the 1992 Consensus, which aligns with the Taiwanese constitution, but opposed the One Country Two Systems framework and vehemently opposed Taiwanese independence under the current administration.

If the differences in cross-strait relations positions between the KMT and the DPP had narrowed, on what basis did Taiwanese voters cast their votes for each party? First, Taiwanese voters with a Taiwanese identity may not have trusted the KMT's policy changes. Second, they might have voted based on the differences in domestic policy issues proposed by each party. Third, they could have voted according to their partisanship. In this case, while there may be variations in the intensity of partisanship, the voting was fundamentally driven by partisan allegiance. Excluding fake news that triggers Taiwanese identity regarding cross-strait relations, a hypothesis associated with partisanship-evoking fake news can be formulated as follows.

- H2: Domestic political fake news is likely to worsen perceptions of the opposing party.
- H2a: When KMT supporters are exposed to domestic political fake news, their perception of the DPP is likely to deteriorate.
- H2b: When DPP supporters are exposed to domestic political fake news, their perception of the KMT is likely to deteriorate.

Bayesian Learning Hypothesis

Taiwan operates under a dual-party system with the Kuomintang (KMT) and the Democratic Progressive Party (DPP). However, according to party preference surveys, independents make up the largest proportion at 56%. Additionally, the 2024 presidential

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the Constitution of the ROC - Making a bold statement, declaring alignment with Deep Blue voters「stand together」." China Daily. July 4. https://www.chinatimes.com/newspapers/20230704000325-260118?chdtv. (Mandarin Version)

election witnessed a surge in the third party, the Taiwan People's Party (TPP). The TPP primarily garnered support from the 2030 generation[4], who have been disappointed with the policies of both the KMT and DPP regarding livelihood issues. As a new party, the TPP attracted individuals less swayed by partisanship in the dual-party system, thus likely to pursue accuracy through Bayesian updating when encountering fake news rather than partisan-motivated reasoning. Moreover, Bayesian updating is expected to occur among independent voters as well.

- H3: Independent voters, when exposed to domestic political fake news, are likely to reach accurate information through Bayesian updating.
- H3a: Supporters of the TPP, when exposed to domestic political fake news, are likely to reach accurate information through Bayesian updating.

However, an alternative prediction is also possible. Supporters of the TPP may back the party due to disillusionment with the traditional two-party system. Thus, exposure to fake news could deepen their negative perceptions of both the KMT and the DPP. Consequently, the following hypothesis can be proposed:

- H3b: When TPP supporters are exposed to domestic political fake news, their perception of the KMT is likely to deteriorate.
- H3b: When TPP supporters are exposed to domestic political fake news, their perception of the DPP is likely to deteriorate.

Mixed Effects of Two Identities

When national identity and partisanship are both salient, another scenario can be

[4] I will explain what this generation is.

considered. For instance, during the presidential election period in Taiwan, fake videos of President Tsai Ing-wen were circulated under the title "The Secret History of Tsai Ing-wen."[5] Although this news is unrelated to China, voters with DPP partisanship may speculate the source of this news to be China, thereby triggering national identity sentiments. If Taiwanese national identities are stronger among KMT supporters, they may attribute the source of this news to China, potentially leading to a backlash effect. However, if partisanship is stronger, there is a possibility that these KMT supporters might accept this fake news. Therefore, the following potential hypotheses can be formulated:

- H4: If Taiwanese national identity is salient, the negative perception of the CCP among DPP supporters increases.
- H4a: If Taiwanese national identities are salient, the negative perception of the CCP among KMT supporters increases.
- H4b: If partisanship is salient, KMT partisans are more likely to accept fake news, regardless of its source.

Finally, people are more likely to believe information as true, regardless of its actual validity, if they are exposed to it frequently (Hasher, Goldstein, and Toppino 1977). This applies to the information processing mechanism of individuals regarding fake news as well (Pillai and Fazio 2021). While topics that evoke identity may not always garner significant attention or controversy, and despite addressing the same subject matter, the degree to which they evoke identity may vary. Fake news on such topics might have less salience compared to other news items that evoke identity. Consequently, individuals' information processing mechanisms may differ depending on the saliency of the fake news.

[5] Tsai and Chin. "China is posting fake videos of president: sources." Taipei Times, January 11, 2024. https://www.taipeitimes.com/News/front/archives/2024/01/11/2003811930.

- H5: Citizens exposed to low-salience fake news will pursue accuracy through Bayesian learning regardless of the topic's type.

4. Research Design

The research design is divided into two phases. In the first phase, I test whether polarization, Bayesian updating, and the backlash effect are also valid in Taiwanese contexts (H1 through H3). In the second phase, I test the effects of mixed identities after confirming the effects of each information process mechanism (H4 and H5).

1) Independent Variables

Independent variables are type of topics, saliency of topics, and Taiwanese people's identities. I conduct topic modeling to identify which fake news triggers certain identities. I compared topics generated by LDA and NMF algorithms and select the algorithm that provides the most intuitive topic classification. Then, I positioned the classified topics on the quadrants shown in Figure 1. The saliency of each topic is measured using the Herfindahl-Hirschman Index (HHI). Lastly, I am planning to collect panel data to measure Taiwanese people's identities.

a. Topic Modeling

To conduct topic modeling, I analyzed data from the six months leading up to the Taiwanese presidential election on January 13, 2024.[6] This dataset comprises 332 entries of fake news and their corresponding refutations, sourced from the Taiwan FactCheck Center. For the topic modeling, I employed two algorithms: Latent Dirichlet Allocation (LDA) and Non-

[6] This analysis served as a preliminary study due to the time-consuming nature of calculating coherence values, which necessitated the use of data from only the past six months in the term paper. In the future development phase, I plan to conduct topic modeling on four years of fake news data.

negative Matrix Factorization (NMF). As a result of comparing the topic categorization of the two algorithms, NMF classified topics more intuitively. Therefore, this study will proceed with analyses based on the NMF model. Table 1 show the results of performing NMF with six topics. Details on the topic modeling can be found in the Appendix.

Table 2. NMF Top 20 Keywords

Topic	Keywords
International relations	China, USA, Reporting, Media, Content, Forgery, Ukraine, Footage, President, Imagery, Reality, Xi Jinping, Platforms, Taiwan, Review, Zelensky, Russia, Pictures, Fabrication, Related
Fukushima Treated Water Release	Japan, Wastewater, Emissions, Fukushima, Nuclear Power Plant, Treatment, Radiation, Imagery, Landing, Seafood, Sardines, Marine Life, Mutation, Large Amount, Experts, Incident, Fish Schools, Misappropriation, Irrelevant
Israel-Hamas War	Israel, Israel-Palestine, Conflict, Hamas, Incident, War, Gaza, Raid, Footage, Irrelevant, Egypt, Palestine, Misappropriation, Imagery, Outbreak, social media, Attacked by Hamas, Original, Syria, Platforms
Imported Egg Controversies	Taiwan, Eggs, Public, Imports, Updates, Experts, Noted, Incident, Stated, Report, Information, Use, Mistake, Statement, Part, Vaccine, Occurrence, If There Is, Synchronized, Misunderstanding
Earthquake at Noto Province, Japan	Earthquake, Japan, Noto, Tsunami, Artificial, Peninsula, Triggered, Ishikawa Prefecture, Footage, New Year's Day, Occurrence, Alleged, Irrelevant, Creation, Misappropriation, social media, Location of Occurrence, Upon Investigation, North Korea, Incident
Election rigging	Voting, Election Services, Vote Counting, Ballot Stuffing, Voting Places, Ballots, Election Commission, Personnel, Public, Election Committee, Legislators, Vote Counts, Vote Counting, Democratic Progressive Party, Cardboard Boxes, Poll Watching, Calling Votes, Elections, Rumors, Presidential Election

b. Topic Saliency

I used the Herfindahl-Hirschman Index (HHI) to determine the saliency of the fake news categories I had previously classified. While the HHI is typically employed to gauge market concentration within an industry, it can also be adapted to measure the saliency of specific issues. To calculate the saliency of each topic, I first computed the topic-specific weights (w_j) for each document i, then squared the sum of these document-specific weights for each topic.

I divided This value by the square of the total sum of the topic weights to normalize it. The formula is represented as follows:

$$Topic_{_HHI} = \frac{(\sum_{i=1}^{D} w_{ij})^2}{\sum {w_j}^2}$$

The computed HHI values for the six topics are presented in Table 3. The threshold for determining the saliency of topics was set at 0.05.

Table 3. Fake News Topic Saliency

Topic	Weighted Sum for each topic	Saliency
International Relations	0.264	Salient
Fukushima Treated Water Release	0.095	Salient
Israel-Hamas War	0.193	Salient
Imported Egg Controversies	0.335	Salient
Earthquake at Noto Province, Japan	0.037	Not Salient
Election rigging	0.076	Salient

c. Positioning Each Topic within a Quadrant

International Relations: This topic addresses various international relations issues, including the war in Ukraine, as well as tensions between the U.S. and China. It also covers provocative actions by Chinese maritime surveillance vessels in the Taiwan Strait. Taiwanese identity is intertwined with a desire to be recognized not as part of China, but as Taiwan in its own right on the international stage. Thus, I placed this issue, which significantly stirs Taiwanese nationality, outside the second quadrant of salient demarcation due to its prominence.

Fukushima Treated Water Release: This topic pertains to the fake news regarding Japan's discharge of nuclear-contaminated water. As this topic is not associated with China, it does not strongly invoke Taiwanese identity. In contrast, during Japan's release of radioactive water in August 2023, the Democratic Progressive Party (DPP) adopted a reserved stance rather than a

strong protest. According to an August survey by the Taiwan Public Opinion Foundation (TPOF), 63.8% of Taiwanese respondents believed it to be dangerous, while only 31.9% thought it was not dangerous. The Kuomintang (KMT) strongly condemned the DPP government's response.[7] This topic elicits partisanship but invokes weak nationality, so I placed this issue outside the fourth quadrant of salient demarcation.

Isarael-Hamas War: The year 2023 marked a deterioration in the Middle East situation due to Hamas's attacks on Israel and the subsequent Israeli retaliation. This period saw a high incidence of misinformation, indicating the salience of recent developments. Additionally, an online survey revealed that nearly 61% of respondents believe that the escalation of the Israel-Hamas conflict could pose a significant threat to Taiwan's security (Hsiao 2023). Consequently, I have positioned this issue as a topic that provokes Taiwanese nationality and placed it outside the second quadrant of saliency demarcation.

Imported Egg Controversies: This topic involves ongoing controversies about the safety of imported eggs, including issues related to fake news about eggs, food safety, and imported ingredients. Since this topic is not related to partisanship or nationality and has high saliency, I positioned this issue in the outside of the saliency demarcation of the third quadrant.

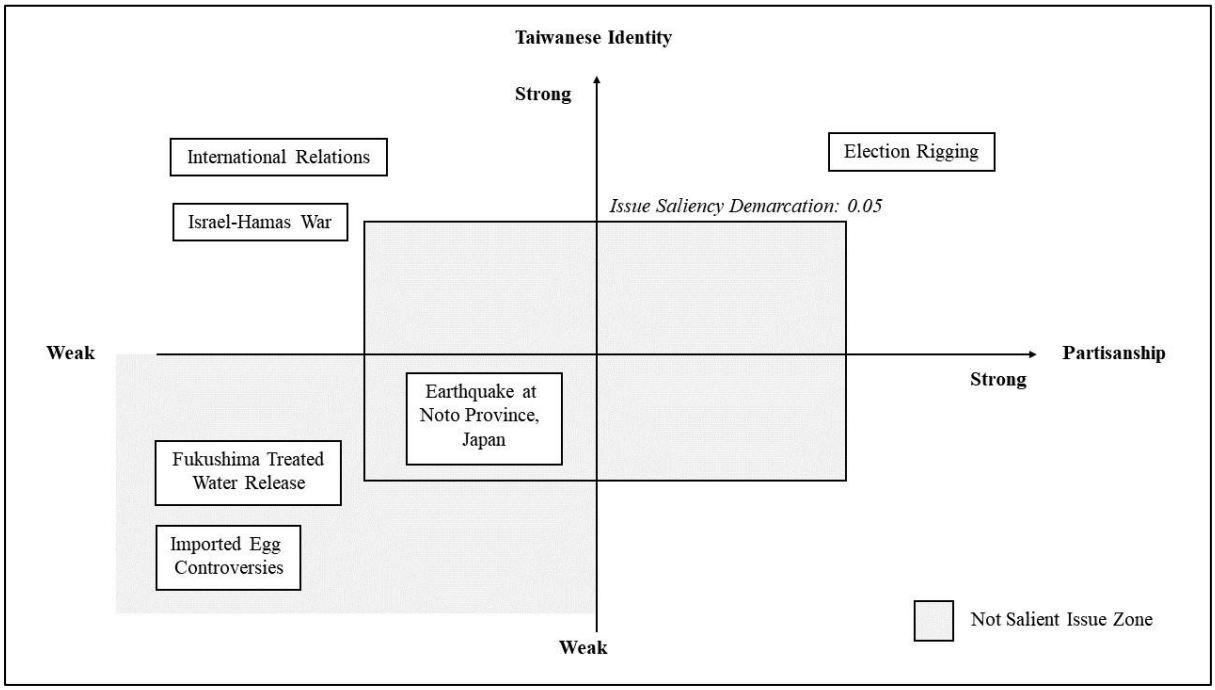
Earthquake at Noto Province, Japan: This topic concerns fake news about an earthquake in Japan's Noto province. While also related to safety, it does not evoke nationality or partisanship. Moreover, this issue has low saliency. Therefore, I placed this issue in the third quadrant and within the saliency demarcation.

Election rigging: This topic is related to fake news concerning the 2024 Taiwanese

[7] Zhou, Yuxiang and Xu Fangci. 2023. "Nuclear Wastewater May Pollute Taiwanese Aquatic Products; Japanese Media Person Surprised: The Taiwan Government Has Not Made Any Noise." China Times, August 8, 2023. Accessed April 23, 2024. https://www.chinatimes.com/realtimenews/20230828003134-260407?chdtv.

presidential election and manipulation of elections in general. Elections are associated with partisanship and, due to concerns over China's interference, strongly evoke Taiwanese nationality. Therefore, this topic has been placed in the first quadrant, outside the saliency demarcation.

Figure 2. Position of each topic



2) Dependent Variables

I explored datasets from various institutions, such as the Election Study Center (ESC) at National Chengchi University, Taiwan Public Opinion Foundation (TPOF), Taiwan Election and Democratization Study (TEDS), Central Election Commission (CEC), Survey Research Data Archive (SRDA), and Taiwan National Security Studies (TNSS) at Duke University. I am still waiting for their response.

3) Model

Since I plan to use panel data, I will apply fixed effect model. Fixed effect model has several advantages for dealing with panel data. First, this model can mitigate endogeneity

issues by eliminating unobserved, time-variant characteristics. I expect this model can capture the change of people's perception by fake news over time. Second, this model can control individual-specific traits such as education, gender, income.

I received feedback suggesting that a survey experiment would be ideal for conducting this research. However, at my current stage, this is not feasible. Therefore, I intend to utilize panel survey data to gather information on Taiwanese nationality, partisanship, and feeling thermometers. I have requested access to several potential panel survey datasets from the Survey Research Data Archive and am currently awaiting a response.

5. Future Research Plan

Fundamentally, it is necessary to conduct a survey experiment to capture changes in the political perceptions of Taiwanese people. Although I requested data from the institutions mentioned earlier, the questions in existing panel datasets do not address the impact of fake news, making it difficult to directly assess its effects. Additionally, the survey data from commonly used sources like the Election Study Center (ESC) and the Taiwan National Security Survey (TNSS) are not panel data, limiting their ability to capture changes in public perception. Currently, I am considering a factorial design.

Furthermore, topics corresponding to the fourth quadrant did not appear in the previous topic modeling because the analysis was only conducted on six months of fake news data. Therefore, I will re-run topic modeling with four years of data. Lastly, it is necessary to measure the saliency of topics over time.

Bibliography

English

Achen, Christopher, and T.Y. Wang. 2017. *The Taiwan Voter*. University of Michigan Press.

Backes, Oliver, and Andrew Swab. 2019. "Cognitive Warfare: The Russian Threat to Election Integrity in the Baltic States."

Bauer, Fin, and Kimberly L. Wilson. 2022. "Reactions to China-Linked Fake News: Experimental Evidence from Taiwan." *The China Quarterly* 249: 21–46. doi:10.1017/S030574102100134X.

Beauchamp-Mustafaga, Nathan. 2023. *Chinese Next-Generation Psychological Warfare: The Military Applications of Emerging Technologies and Implications for the United States*. RAND Corporation. https://www.rand.org/pubs/research_reports/RRA853-1.html (February 16, 2024).

Bernal, Alonso, Cameron Carter, Ishpreet Singh, Kathy Cao, and Olivia Madreperla. 2020. "Cognitive Warfare: An Attack on Truth and Thought." https://innovationhub-act.org/wp-content/uploads/2023/12/Cognitive-Warfare.pdf (February 9, 2024).

Boyer, Ming M, Loes Aaldering, and Sophie Lecheler. 2022. "Motivated Reasoning in Identity Politics: Group Status as a Moderator of Political Motivations." *Political Studies* 70(2): 385–401. doi:10.1177/0032321720964667.

Charness, Gary, and Chetan Dave. 2017. "Confirmation Bias with Motivated Beliefs." *Games and Economic Behavior* 104: 1–23. doi:10.1016/j.geb.2017.02.015.

Druckman, James N., and Mary C. McGrath. 2019. "The Evidence for Motivated Reasoning in Climate Change Preference Formation." *Nature Climate Change* 9(2): 111–19. doi:10.1038/s41558-018-0360-1.

Gaines, Brian J., James H. Kuklinski, Paul J. Quirk, Buddy Peyton, and Jay Verkuilen. 2007. "Same Facts, Different Interpretations: Partisan Motivation and Opinion on Iraq." *The Journal of Politics* 69(4): 957–74. doi:10.1111/j.1468-2508.2007.00601.x.

Greene, Derek, and James P. Cross. 2017. "Exploring the Political Agenda of the European Parliament Using a Dynamic Topic Modeling Approach." *Political Analysis* 25(1): 77–94. doi:10.1017/pan.2016.7.

Guess, Andrew, and Alexander Coppock. 2020. "Does Counter-Attitudinal Information Cause Backlash? Results from Three Large Survey Experiments." *British Journal of Political Science* 50(4): 1497–1515. doi:10.1017/S0007123418000327.

Hasher, Lynn, David Goldstein, and Thomas Toppino. 1977. "Frequency and the Conference of Referential Validity." *Journal of Verbal Learning and Verbal Behavior* 16(1): 107–12. doi:10.1016/S0022-5371(77)80012-1.

Hill, Seth J. 2017. "Learning Together Slowly: Bayesian Learning about Political Facts." *The Journal of Politics* 79(4): 1403–18. doi:10.1086/692739.

Ho, Yi-Chun (Chad), Junjie Wu, and Yong Tan. 2017. "Disconfirmation Effect on Online Rating Behavior: A Structural Model." *Information Systems Research* 28(3): 626–42. doi:10.1287/isre.2017.0694.

Howard, Philip N, Bharath Ganesh, Dimitra Liotsiou, John Kelly, and Camille François. 2019. "The IRA, Social Media and Political Polarization in the United States, 2012-2018." *Select Committee on Intelligence United States Senate*.

Hsiao, Russell. 2023. "Assessing the Knock-On Effects of Taiwanese Perceptions toward the Isra- El-Hamas War." *Global Taiwan Brief* 8(21): 1–13.

Jost, John T., Delia S. Baldassarri, and James N. Druckman. 2022. “Cognitive–Motivational Mechanisms of Political Polarization in Social-Communicative Contexts.” *Nature Reviews Psychology* 1(10): 560–76. doi:10.1038/s44159-022-00093-5.

Lin, Bonny, Cristina L. Garafola, Bruce McClintock, Jonah Blank, Jeffrey W. Hornung, Karen Schwindt, Jennifer D. P. Moroney, et al. 2022. *A New Framework for Understanding and Countering China's Gray Zone Tactics*. RAND Corporation. https://www.rand.org/pubs/research_briefs/RBA594-1.html (January 27, 2024).

McKay, Spencer, and Chris Tenove. 2021. “Disinformation as a Threat to Deliberative Democracy.” *Political Research Quarterly* 74(3): 703–17. doi:10.1177/1065912920938143.

Pennycook, Gordon, Tyrone D. Cannon, and David G. Rand. 2018. “Prior Exposure Increases Perceived Accuracy of Fake News.” *Journal of Experimental Psychology: General* 147(12): 1865–80. doi:10.1037/xge0000465.

Pillai, Raunak M., and Lisa K. Fazio. 2021. “The Effects of Repeating False and Misleading Information on Belief.” *WIREs Cognitive Science* 12(6): e1573. doi:10.1002/wcs.1573.

Sota, CD de La, and Amory Gethin. 2021. “Inequality, Identity, and the Structure of Political Cleavages in South Korea, Taiwan, and Hong Kong, 1996-2016.”

Taber, Charles S., and Milton Lodge. 2006. “Motivated Skepticism in the Evaluation of Political Beliefs.” *American Journal of Political Science* 50(3): 755–69.

Tappin, Ben M., Gordon Pennycook, and David G. Rand. 2020. “Bayesian or Biased? Analytic Thinking and Political Belief Updating.” *Cognition* 204: 104375. doi:10.1016/j.cognition.2020.104375.

Unsworth, Kerrie L., and Kelly S. Fielding. 2014. “It's Political: How the Salience of One's Political Identity Changes Climate Change Beliefs and Policy Support.” *Global Environmental Change* 27: 131–37. doi:10.1016/j.gloenvcha.2014.05.002.

Mandarin

Zhou, Yuxiang and Xu Fangci. 2023. "Nuclear Wastewater May Pollute Taiwanese Aquatic Products; Japanese Media Person Surprised: The Taiwan Government Has Not Made Any Noise." China Times, August 8, 2023. Accessed April 23, 2024. https://www.chinatimes.com/realtimenews/20230828003134-260407?chdtv.

Appendix

1. Topic Modeling

1) Data Collection and Preprocessing

A. Data collection

The Taiwan FactCheck Center collects various pieces of fake news circulating online and, in the media, conducting analyses to verify their authenticity. The categorization of fake news includes classifications such as fake, partially fake, fact clarification, and insufficient evidence. Fact-check articles are structured with a summary of the verification, background information, and a detailed verification section. The verification summary includes details about the fake news and its specific refutations, which is why I chose to scrape the verification summaries for my analysis. In this term paper, I analyzed a total of 332 fact-check articles from July 2023 to January 2024.

B. Preprocessing

Unlike English, Chinese does not require stemming or lemmatization. However, due to the characteristics of Chinese characters that express meanings, a single character can be an independent word or combine with other characters to form a word with a specific meaning. For example, 部 (bu) can signify a quantity on its own, but it also combines with other characters to form new words, as in 國防部 (Department of Defense, Guo Fang Bu). Therefore, during tokenization, there can be instances where 部 is recognized as a separate word, resulting in 國防 (National Defense) and 部 (Department or Province) being tokenized as two separate words. To address such issues, I included custom words in the package's dictionary during the tokenization process, helping prevent the unnecessary segmentation of words. I employed a dictionary from Jieba, a Chinese tokenization software available on

GitHub, for the customized dictionary. This involved converting simplified Chinese characters into traditional Chinese characters. The customized dictionary contains a total of 349,773 words.

For the stopwords, I combined Jieba's default stopwords list with the TCSP (Traditional Chinese Specialized Project) stopwords list, which is tailored for analyzing traditional Chinese text. After an initial tokenization, I identified and added any remaining unnecessary words that were not removed, resulting in a comprehensive list of 1,165 stopwords applied to the analysis.

2) Optimal Number of Topics

For the topic modeling, I employed two algorithms: Latent Dirichlet Allocation (LDA) and Non-negative Matrix Factorization (NMF). LDA is widely used for topic modeling; however, I also incorporated NMF because it has been shown to highlight niche topics more effectively, providing a richer, more intuitive analysis (Greene and Cross 2017).

To determine the optimal number of topics, I first calculated coherence values across a broad range. I used the coherence values C_v for topic counts $k_i = 2^i, for\ i \in \{1,2,\dots,7\}$. For the training of LDA and NMF models, I employed the default parameters and conducted the training 20 times to ensure stable results, taking the average of these runs. The coherence values across this broad range are shown in Figure A1.

As indicated in Figure A1, the optimal number of topics is found to be between 1 and 10. Therefore, I narrowed the range to $k_i = i, for\ i \in \{1,2,\dots,10\}$ and repeated the process to calculate the coherence values C_v. The results of this refined calculation are presented in Figure A2. From these calculations, it was determined that the optimal number of topics is 6. Subsequently, I proceeded with NMF and LDA modeling using this optimal topic number.

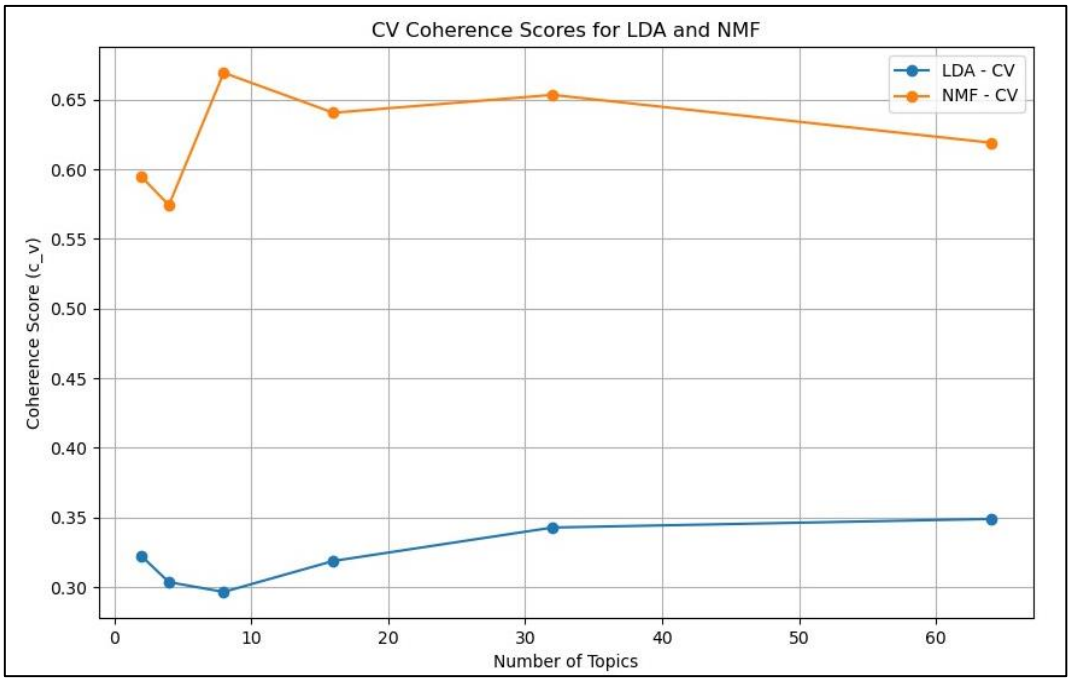


Figure A1. Broad Range C_v values for LDA and NMF

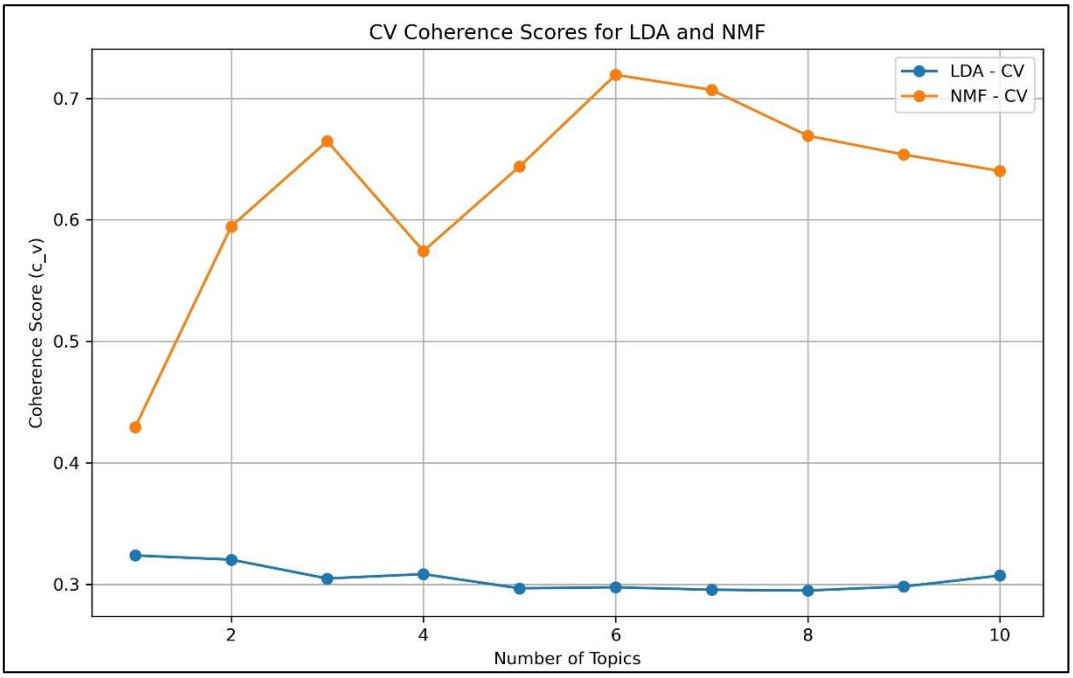


Figure A2. Narrow Range C_v values for LDA and NMF

3) Comparing LDA and NMF

After running LDA and NMF models, I identified the top 20 keywords, which are listed in Tables A1 to A4. In the LDA model, to extract the top 20 keywords while considering the saliency and discriminative power of words within each topic, I adjusted the saliency parameter

(λ) with respective weights of 0.4, 0.6, and 1.0 for extracting the keywords.

From the keywords of the LDA topics in Table 1, it is evident that the topics are generally underdeveloped. The keywords marked in red all represent potential conceptually spurious or multi-context words. Additionally, increasing the saliency parameter tends to make the topic's keywords more abstract, making it difficult to grasp the concept. In contrast, an examination of the topic categorization and keywords of NMF reveals that the keywords effectively reflect the concept of each topic and demonstrate consistency.

Table A1. LDA Top 20 Keywords, λ = 0.4

Topic	Keywords
Topic 1	Voting, Flood, Pond, Temporary Stopping, New Party, Red line, Drugs, (Ballot) Box, Amendment, Ballot, Draft, Ko Wen-je, Chao Xiao-kang, Recording the Number of Votes, Cardboard box, Selection, Mayor, Meter, Rocket, Basketball Court
Topic 2	Forgery, Egg, Fungus, Content, Mold, Xi Jinping, import, sound, media, clam, Ukraine, Mistake, AI, Hawaii, China, Refugee, Pneumonia, Recruitment, Question, Original
Topic 3	Emissions, Wastewater, Nuclear, Fukushima, Tritium, Treatment, Japan, Radiation, Debt raising, nuclear power plant, Religion, Transfer, Standard, Expert, Li Keqiang, Promote, South Korea, Passport, Seawater, Seafood
Topic 4	Hornet, Taiwan, Mobilization, Time, Announcement, Tornado, Egypt, document, Scholar, Protest, Plan, Convene, Surveillance, Blockade, Annette Lu (former vice president), Abandon, United Nations, Incident, Votes, Regulation
Topic 5	AIT, Pork, Election Services, Vote Counting, Current Location, Wheat, Neihu, LINE, Lithuania, Seedless, Dioxin, Language, Land, Inspector, Indonesia, Military Aircraft, Marking, Toilet Paper, Military Academy, Larvae
Topic 6	Israel, Israel-Palestine, Conflict, Earthquake, Hamas, Raid, Dengue Fever, Typhoon, Noto, Attacked by Hamas, Financial Settlement, Gaza, Drill, Decommissioned, Mistake, Misappropriation, War, Syria, Hostage, Military Conflict

Table A2. LDA Top 20 Keywords, λ = 0.6

Topic	Keywords
Topic 1	voting, flood, pond, (ballot) box, temporary parking, ballot, red line, drugs, Ko Wen-je, New Party, legislation amendment, Zhao Xiaokang, draft bill, legislator, recording point, cardboard box, duration, meters, rocket, Russia
Topic 2	content, error, forgery, egg, media, import, China, Taiwan, bacteria, mycoplasma, Xi Jinping,

	platform, Ukraine, social group, original, report, image, indicate, real, point out
Topic 3	emission, wastewater, nuclear, Fukushima, Japan, tritium, treatment, expert, error, point out, nuclear power plant, radiation, debt raising, event, standard, religion, transfer, chicken egg, South Korea, egg
Topic 4	Taiwan, error, event, hornet, report, time, media, USA, plan, information, mobilization, advance notice, mobilization, Egypt, protest, Japan, book, tornado, expert, regulation
Topic 5	Taiwan, AIT, pork, error, electoral affairs, public, image, USA, vote counting, marking, current location, pandemic, ballot opening, wheat, LINE, reporting, appearance, Neihu, Lithuania, image
Topic 6	Israel, Israel-Palestine, conflict, error, earthquake, Hamas, raid, dengue fever, misappropriation, typhoon, image, USA, related, unrelated, war, exercise, Gaza, social group, refer, event

Table A3. LDA Top 20 Keywords, λ = 1.0

Topic	Keywords
Topic 1	voting, error, public, indicates, expert, China, Taiwan, flood, (ballot) box, ballot, Ko Wen-je, legislator, vaccine, Russia, pond, drugs, temporary parking, red line, points out, ballot tampering, open
Topic 2	error, content, media, Taiwan, China, social group, report, forgery, platform, egg, import, points out, image, indicates, original, imagery, Ukraine, unrelated, review, related
Topic 3	emission, wastewater, Japan, error, expert, nuclear, Fukushima, points out, event, treatment, tritium, media, public, content, unrelated, social group, report, egg, China, import
Topic 4	error, Taiwan, event, report, media, USA, expert, Japan, social group, indicates, image, information, China, time, points out, network, occurs, content, use, platform
Topic 5	error, Taiwan, public, image, USA, report, media, content, event, points out, China, imagery, expert, social group, electoral affairs, pork, indicates, appears, information, review
Topic 6	error, Israel, image, USA, Israel-Palestine, social group, conflict, event, earthquake, misappropriation, public, related, unrelated, China, Japan, expert, content, Hamas, points out, review

Table A4. NMF Top 20 Keywords

Topic	Keywords
International relations	China, USA, Reporting, Media, Content, Forgery, Ukraine, Footage, President, Imagery, Reality, Xi Jinping, Platforms, Taiwan, Review, Zelensky, Russia, Pictures, Fabrication, Related
Fukushima Treated Water Release	Japan, Wastewater, Emissions, Fukushima, Nuclear Power Plant, Treatment, Radiation, Imagery, Landing, Seafood, Sardines, Marine Life, Mutation, Large Amount, Experts, Incident, Fish Schools, Misappropriation, Irrelevant
Israel-Hamas War	Israel, Israel-Palestine, Conflict, Hamas, Incident, War, Gaza, Raid, Footage, Irrelevant, Egypt, Palestine, Misappropriation, Imagery, Outbreak, social media, Attacked by Hamas, Original, Syria, Platforms

Imported Egg Controversies	Taiwan, Eggs, Public, Imports, Updates, Experts, Noted, Incident, Stated, Report, Information, Use, Mistake, Statement, Part, Vaccine, Occurrence, If There Is, Synchronized, Misunderstanding
Earthquake at Noto Province, Japan	Earthquake, Japan, Noto, Tsunami, Artificial, Peninsula, Triggered, Ishikawa Prefecture, Footage, New Year's Day, Occurrence, Alleged, Irrelevant, Creation, Misappropriation, social media, Location of Occurrence, Upon Investigation, North Korea, Incident
Election rigging	Voting, Election Services, Vote Counting, Ballot Stuffing, Voting Places, Ballots, Election Commission, Personnel, Public, Election Committee, Legislators, Vote Counts, Vote Counting, Democratic Progressive Party, Cardboard Boxes, Poll Watching, Calling Votes, Elections, Rumors, Presidential Election

4) LDA and NMF Top 20 Keywords (Traditional Mandarin)

Table B1. LDA Top 20 Keywords, λ = 0.4

Topic	Keywords
Topic 1	投票, 滯洪, 池, 臨停, 新黨, 紅線, 毒品, 甌, 修法, 票, 草案, 柯文哲, 趙小康, 記點, 紙箱, 遴選, 市長, 公尺, 火箭, 籃球場
Topic 2	變造, 雞蛋, 菌, 內容, 黴漿, 習近平, 進口, 聲音, 媒體, 文蛤, 烏克蘭, 錯誤, AI, 夏威夷, 中國, 難民, 肺炎, 招募, 提問, 原始
Topic 3	排放, 廢水, 核, 福島, 氚, 處理, 日本, 輻射, 舉債, 核電廠, 宗教, 轉會, 標準, 專家, 李克強, 促, 韓國, 護照, 海水, 海鮮
Topic 4	虎頭蜂, 台灣, 動員, 時間, 預告, 龍捲風, 埃及, 書, 學人, 抗議, 計畫, 召集, 監視器, 封鎖, 呂秀蓮, 放棄, 聯合國, 事件, 票數, 規定
Topic 5	AIT, 豬肉, 選務, 計票, 現址, 小麥, 內湖, LINE, 立陶宛, 無籽, 戴奧辛, 語言, 土地, 監察員, 印尼, 軍機, 標示, 衛生紙, 陸軍官校, 孑孓
Topic 6	以色列, 以巴, 衝突, 地震, 哈瑪斯, 突襲, 登革熱, 颱風, 能登, 遭哈瑪斯, 決算, 加薩, 演習, 除役, 錯誤, 挪用, 戰爭, 敘利亞, 人質, 軍事衝突

Table B2. LDA Top 20 Keywords, λ = 0.6

Topic	Keywords
Topic 1	投票, 滯洪, 池, 甌, 臨停, 票, 紅線, 毒品, 柯文哲, 新黨, 修法, 趙小康, 草案, 立委, 記點, 紙箱, 時長, 公尺, 火箭, 俄羅斯
Topic 2	內容, 錯誤, 變造, 雞蛋, 媒體, 進口, 中國, 台灣, 菌, 黴漿, 習近平, 平台, 烏克蘭, 社群, 原始, 報導, 影像, 表示, 真實, 指出
Topic 3	排放, 廢水, 核, 福島, 日本, 氚, 處理, 專家, 錯誤, 指出, 核電廠, 輻射, 舉債, 事件, 標準, 宗教, 轉會, 雞蛋, 韓國, 蛋
Topic 4	台灣, 錯誤, 事件, 虎頭蜂, 報導, 時間, 媒體, 美國, 計畫, 資訊, 動員, 預告, 動員, 埃及, 抗議, 日本, 書, 龍捲風, 專家, 規定
Topic 5	台灣, AIT, 豬肉, 錯誤, 選務, 民眾, 畫面, 美國, 計票, 標示, 現址, 疫情, 開票, 小麥, LINE, 報道, 出現, 內湖, 立陶宛, 影像

Topic 6	以色列，以巴，衝突，錯誤，地震，哈瑪斯，突襲，登革熱，挪用，颱風，畫面，美國，相關，無關，戰爭，演習，加薩，社群，指稱，事件

Table B3. LDA Top 20 Keywords, λ = 1.0

Topic	Keywords
Topic 1	投票，錯誤，民眾，表示，專家，中國，台灣，滯洪，甌，票，柯文哲，立委，疫苗，俄羅斯，池毒品，臨停，紅線，指出，作票，開放
Topic 2	錯誤，內容，媒體，台灣，中國，社群，報導，變造，平台，雞蛋，進口，指出，畫面，表示，原始，影像，烏克蘭，無關，檢視，相關
Topic 3	排放，廢水，日本，錯誤，專家，核，福島，指出，事件，處理，氚，媒體，民眾，內容，無關，社群，報告，雞蛋，中國，進口
Topic 4	錯誤，台灣，事件，報導，媒體，美國，專家，日本，社群，表示，畫面，資訊，中國，時間，指出，網路，發生，內容，使用，平台
Topic 5	錯誤，台灣，民眾，畫面，美國，報導，媒體，內容，事件，指出，中國，影像，專家，社群，選務，豬肉，表示，出現，資訊，檢視
Topic 6	錯誤，以色列，畫面，美國，以巴，社群，衝突，事件，地震，挪用，民眾，相關，無關，中國，日本，專家，內容，哈瑪斯，指出，檢視

Table B4. NMF Top 20 Keywords

Topic	Keywords
Topic 1	中國，美國，報導，媒體，內容，變造，烏克蘭，畫面，總統，影像，真實，習近平，平台，台灣，檢視，倫斯基，俄羅斯，圖片，虛構，相關
Topic 2	日本，廢水，排放，福島，核電廠，處理，輻射，影像，上岸，海鮮，沙丁魚，海洋生物，巴馬，變異，大量，專家，事件，魚群，挪用，無關
Topic 3	以色列，以巴，衝突，哈瑪斯，事件，戰爭，加薩，突襲，畫面，無關，埃及，巴勒斯坦，挪用，影像，爆發，社群，遭哈瑪斯，原始，敘利亞，平台
Topic 4	台灣，雞蛋，民眾，進口，更新，專家，指出，事件，表示，報告，資訊，使用，錯誤，聲明，部分，疫苗，發生，若有，同步，誤解
Topic 5	地震，日本，能登，海嘯，人工，半島，引發，石川縣，畫面，元旦，發生，指稱，無關，創作，挪用，社群，發生地，經查，北韓，事件
Topic 6	投票，選務，開票，作票，投開票所，選票，中選會，人員，民眾，選委會，立委，票數，計票，民進黨，紙箱，監票，唱票，選舉，謠言，總統大選